

- (a) $\text{Im}(F) = 0$ on Γ .
- (b) $\text{Re}(F)$ has a minimum on Γ at some point z_0 , and this function is non-degenerate in the sense that the second derivative of $\text{Re}(F)$ along Γ is strictly positive at z_0 .

Conditions (a) and (b) imply of course that $F'(z_0) = 0$. If as above, $F''(z_0) \neq 0$, then there are two curves Γ_1 and Γ_2 passing through z_0 which are orthogonal, so that $F|_{\Gamma_i}$ is real for $i = 1, 2$, with $\text{Re}(F)$ restricted to Γ_1 having a minimum at z_0 ; and $\text{Re}(F)$ restricted to Γ_2 having a maximum at z_0 (see Exercise 2 in Chapter 8). We therefore try to deform our original contour of integration to $\Gamma = \Gamma_1$. This approach is usually referred to as the method of **steepest descent**, because at z_0 the function $-\text{Re}(F(z))$ has a saddle point, and starting at this point and following the path of Γ_1 , one has the greatest decrease of this function.

Let us return to our special case, $F(z) = -i(z^3/3 + z)$. We note that

$$\begin{cases} \text{Re}(F) &= x^2y - y^3/3 + y, \\ \text{Im}(F) &= -x^3/3 + xy^2 - x. \end{cases}$$

We observe also that $F'(z) = -i(z^2 + 1)$, so we have two non-real critical points $z_0 = \pm i$ at which $F'(z_0) = 0$. If we choose $z_0 = i$, then the two curves passing through this point where $\text{Im}(F) = 0$ are

$$\Gamma_1 = \{(x, y) : y^2 = x^2/3 + 1\} \quad \text{and} \quad \Gamma_2 = \{(x, y) : x = 0\}.$$

On Γ_2 , the function $\text{Re}(F)$ clearly has a maximum at the point $z_0 = i$, and so we reject this curve. We choose $\Gamma = \Gamma_1$, which is a branch of a hyperbola, and which can be written as $y = (x^2/3 + 1)^{1/2}$; it is asymptotic to the rays $z = re^{i\pi/6}$, and $z = re^{i5\pi/6}$ at infinity. See Figure 4.

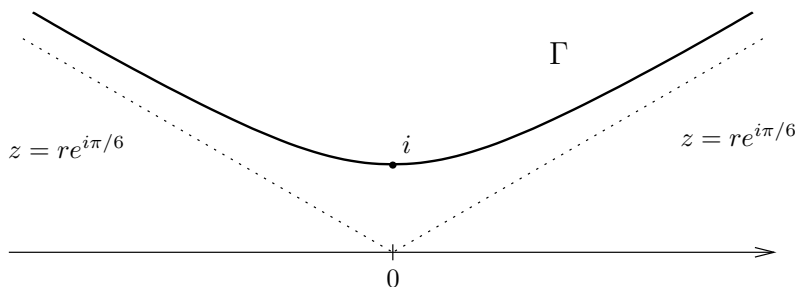


Figure 4. The curve of steepest descent

Next, we see that

$$(21) \quad \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-sF(x)} dx = \frac{1}{2\pi} \int_{\Gamma} e^{-sF(z)} dz.$$

This identity is justified by applying the Cauchy theorem to $e^{-sF(z)}$ on the contour Γ_R that consists of four arc segments: the parts of the real axis and Γ that lie inside the circle of radius R , and the two arcs of this circle joining the axis with Γ . Since in this region $e^{-sF(z)} = O(e^{-cyx^2})$ as $x \rightarrow \pm\infty$, the contributions of the two arcs of the circle are $O(\int_0^\pi e^{-cR^2 \sin \theta} d\theta) = O(1/R)$, and letting $R \rightarrow \infty$ establishes (21).

We now observe that on Γ

$$\Phi(x) = \operatorname{Re}(F) = y(x^2 - y^2/3 + 1) = \left(\frac{8}{9}x^2 + \frac{2}{3}\right)(x^2/3 + 1)^{1/2},$$

since $y^2 = x^2/3 + 1$ there. Also, on Γ we have that $dz = dx + idy = dx + i(x/3)(x^2/3 + 1)^{-1/2}dx$. Thus,

$$(22) \quad \frac{1}{2\pi} \int_{\Gamma} e^{-sF(z)} dz = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-s\Phi(x)} dx,$$

in view of the fact that $\Phi(x)$ is even, while $x(x^2/3 + 1)^{-1/2}$ is odd.

We note next that since $(1 + u)^{1/2} = 1 + u/2 + O(u^2)$ as $u \rightarrow 0$,

$$\Phi(x) = \left(\frac{8}{9}x^2 + \frac{2}{3}\right) + \frac{2}{3} \frac{1}{2} \frac{x^2}{3} + O(x^4) = x^2 + \frac{2}{3} + O(x^4),$$

and so $\Phi''(0) = 2$. We now apply Proposition 2.1 to estimate the main part of the right-hand side of (22), by

$$\frac{1}{2\pi} \int_{-c}^c e^{-s\Phi(x)} dx,$$

where c is a small positive constant. Since $\Phi(0) = 2/3$, $\Phi''(0) = 2$, and $\psi(0) = 1/2\pi$, we obtain that this term contributes

$$e^{-\frac{2}{3}s} \left[\frac{1}{2\pi^{1/2}} \frac{1}{s^{1/2}} + O\left(\frac{1}{s}\right) \right].$$

The term $\int_c^\infty e^{-s\Phi(x)} dx$ is dominated by $e^{-2s/3} \int_c^\infty e^{-c_1sx^2} dx$, which is $O(e^{-2s/3}e^{-\delta s})$ for some $\delta > 0$, as soon as $c > 0$. A similar estimate holds for $\int_{-\infty}^{-c} e^{-s\Phi(x)} dx$. Altogether, then,

$$I_+(s) = e^{-\frac{2}{3}s} \left[\frac{1}{2\pi^{1/2}} \frac{1}{s^{1/2}} + O\left(\frac{1}{s}\right) \right] \quad \text{as } s \rightarrow \infty,$$

and this gives the desired asymptotic (ii) for the Airy function.

4 The partition function

Our last illustration of the techniques developed in this appendix is in their application to the partition function $p(n)$, which was discussed in Chapter 10. We derive for it the main term of the remarkable asymptotic formula of Hardy-Ramanujan.

Theorem 4.1 *If p denotes the partition function, then*

$$(i) \quad p(n) \sim \frac{1}{4\sqrt{3}n} e^{Kn^{1/2}} \text{ as } n \rightarrow \infty, \text{ where } K = \pi\sqrt{\frac{2}{3}}.$$

(ii) *A much more precise assertion is that*

$$p(n) = \frac{1}{2\pi\sqrt{2}} \frac{d}{dn} \left(\frac{e^{K(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}} \right) + O(e^{\frac{K}{2}n^{1/2}}).$$

Note. Observe that $(n - \frac{1}{24})^{1/2} - n^{1/2} = O(n^{-1/2})$, by the mean-value theorem; hence $e^{K(n-\frac{1}{24})^{1/2}} = e^{Kn^{1/2}}(1 + O(n^{-1/2}))$, thus $e^{K(n-\frac{1}{24})^{1/2}} \sim e^{Kn^{1/2}}$, as $n \rightarrow \infty$. Of course, clearly $(n - \frac{1}{24})^{1/2} \sim n^{1/2}$, and in particular (ii) implies (i).

We shall discuss first, in a more general setting, how we might derive the asymptotic behavior of a sequence $\{F_n\}$ from the analytic properties of its generating function $F(w) = \sum_{n=0}^{\infty} F_n w^n$. Assuming for the sake of simplicity that $\sum F_n w^n$ has the unit disc as its disc of convergence, we can set forth the following heuristic principle: the asymptotic behavior of F_n is determined by the location and nature of the “singularities” of F on the unit circle, and the contribution to the asymptotic formula due to each singularity corresponds in magnitude to the “order” of that singularity.

A very simple example in which this principle is unambiguous and can be verified occurs when F is meromorphic in a larger disc, but has only one singularity on the circle, a pole of order r at the point $w = 1$. Then there is a polynomial P of degree $r - 1$ so that $F_n = P(n) + O(e^{-\epsilon n})$ as $n \rightarrow \infty$, for some $\epsilon > 0$. In fact, $\sum_{n=0}^{\infty} P(n)w^n$ is a good approximation to $F(w)$ near $w = 1$; it is the principal part of the pole of F . (See also Problem 4.)

For the partition function the analysis is not as simple as this example, but the principle stated above is still applicable when properly interpreted. To this task we now turn.

We recall the formula

$$\sum_{n=0}^{\infty} p(n) w^n = \prod_{n=1}^{\infty} \frac{1}{1 - w^n},$$

established in Theorem 2.1, Chapter 10. This identity implies that the generating function is holomorphic in the unit disc. In what follows, it will be convenient to pass from the unit disc to the upper half-plane by writing $w = e^{2\pi iz}$, $z = x + iy$, and taking $y > 0$. We therefore have

$$\sum_{n=0}^{\infty} p(n) e^{2\pi inz} = f(z),$$

with

$$f(z) = \prod_{n=1}^{\infty} \frac{1}{1 - e^{2\pi inz}},$$

and

$$(23) \quad p(n) = \int_{\gamma} f(z) e^{-2\pi inz} dz.$$

Here γ is the segment in the upper half-plane joining $-1/2 + i\delta$ to $1/2 + i\delta$, with $\delta > 0$; the height δ will be fixed later in terms of n .

To proceed further, we look first at where the main contribution to the integral (23) might be, in terms of the relative size of $f(x + iy)$, as $y \rightarrow 0$. Notice that f is largest near $z = 0$. This is because $|f(x + iy)| \leq f(iy)$, and moreover $f(iy)$ increases as y decreases, in view of the fact that the coefficients $p(n)$ are positive. Alternatively, we observe that each factor $1 - e^{2\pi inz}$, appearing in the product for f , vanishes as $z \rightarrow 0$, but the same is true for any other point (mod 1) on the real axis. Thus in analogy with the simple example considered above, we seek an elementary function f_1 , which has much the same behavior as f at $z = 0$, and try to replace f by f_1 in (23).

It is here that we are very fortunate, because the generating function is just a variant of the Dedekind eta function,

$$\eta(z) = e^{i\pi z/12} \prod_{n=1}^{\infty} (1 - e^{2\pi inz}).$$

From this, it is obvious that

$$f(z) = e^{\frac{i\pi z}{12}} (\eta(z))^{-1}.$$

(Incidentally, the fraction $1/12$ arising above will explain the occurrence of the fraction $1/24$ in the asymptotic formula for $p(n)$.)

Since η satisfies the functional equation $\eta(-1/z) = \sqrt{z/i} \eta(z)$ (see Proposition 1.9 in Chapter 10), it follows that

$$(24) \quad f(z) = \sqrt{z/i} e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}} f(-1/z).$$

Notice also that if z is appropriately restricted and $z \rightarrow 0$, then $\text{Im}(-1/z) \rightarrow \infty$, from which it follows that $f(-1/z) \rightarrow 1$ rapidly, because

$$(25) \quad f(z) = 1 + O(e^{-2\pi y}), \quad z = x + iy, y \geq 1.$$

Thus it is natural to choose $f_1(z) = \sqrt{z/i} e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}}$ as the function which approximates well the generating function $f(z)$ (at $z = 0$), and write (because of (24))

$$p(n) = p_1(n) + E(n),$$

with

$$\begin{cases} p_1(n) &= \int_{\gamma} \sqrt{z/i} e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}} e^{-2\pi i n z} dz, \\ E(n) &= \int_{\gamma} \sqrt{z/i} e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}} e^{-2\pi i n z} (f(-1/z) - 1) dz. \end{cases}$$

We first take care of the error term $E(n)$, and in doing so we specify γ by choosing its height in terms of n . In estimating $E(n)$ we replace its integrand by its absolute value and note that if $z \in \gamma$, then

$$(26) \quad \left| \sqrt{z/i} e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}} e^{-2\pi i n z} \right| \leq c e^{2\pi n \delta} e^{\frac{\pi}{12} \frac{\delta}{\delta^2 + x^2}},$$

since $z = x + iy$, and $\text{Re}(i/z) = \delta/(\delta^2 + x^2)$.

On the other hand, we can make two estimates for $f(-1/z) - 1$. The first arises from (25) by replacing z by $-1/z$, and gives

$$(27) \quad |f(-1/z) - 1| \leq c e^{-2\pi \frac{\delta}{\delta^2 + x^2}} \quad \text{if } \frac{\delta}{\delta^2 + x^2} \geq 1.$$

For the second, we observe that $|f(z)| \leq f(iy) \leq C e^{\frac{\pi}{12y}}$, when $y \leq 1$, because of the functional equation (24), and hence

$$(28) \quad |f(-1/z) - 1| \leq O\left(e^{\frac{\pi}{12} \frac{\delta^2 + x^2}{\delta}}\right) = O\left(e^{\frac{\pi}{48\delta}}\right)$$

if $\frac{\delta}{\delta^2 + x^2} \leq 1$, since $|x| \leq 1/2$.

Therefore in the integral defining $E(n)$ we use (26) and (27) when $\frac{\delta}{\delta^2+x^2} \geq 1$, and (26) and (28) when $\frac{\delta}{\delta^2+x^2} \leq 1$. The first leads to a contribution of $O(e^{2\pi n\delta})$, since $2\pi > \pi/12$. The second gives a contribution of $O(e^{2\pi n\delta} e^{\frac{\pi}{48\delta}})$. Hence $E(n) = O(e^{2\pi n\delta} e^{\frac{\pi}{48\delta}})$, and we choose δ so as to minimize the right-hand side, that is, $2\pi n\delta = \frac{\pi}{48\delta}$; this means we take $\delta = \frac{1}{4\sqrt{6}n^{1/2}}$, and we get

$$E(n) = O\left(e^{\frac{4\pi}{4\sqrt{6}}n^{1/2}}\right) = O\left(e^{\frac{K}{2}n^{1/2}}\right),$$

which is the desired size of the error term.

We turn to the main term $p_1(n)$. To simplify later calculations we “improve” the contour γ by adding to it two small end-segments; these are the segment joining $-1/2$ to $-1/2 + i\delta$ and that joining $1/2 + i\delta$ to $1/2$. We call this new contour γ' (see Figure 5).

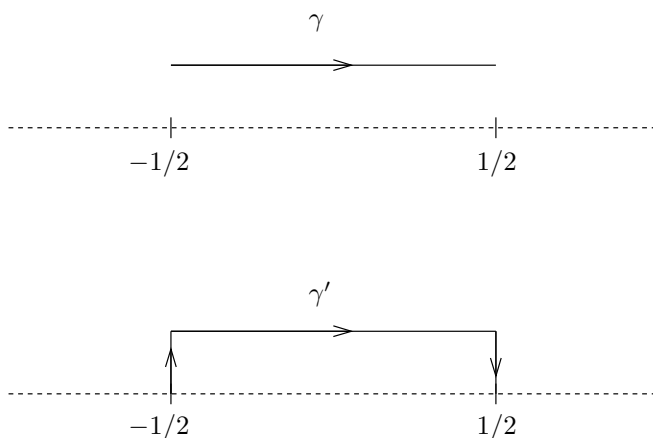


Figure 5. γ and the improved contour γ'

Notice that since $\sqrt{z/i} e^{\frac{i\pi}{12z}}$ is $O(1)$ on the two added segments (for the integral defining $p_1(n)$), the modification contributes $O(e^{2\pi n\delta}) = O(e^{\frac{2\pi}{4\sqrt{6}}n^{1/2}}) = O(e^{\frac{K}{4}n^{1/2}})$, which is even smaller than the allowed error, and therefore can be incorporated in $E(n)$. So without introducing further notation we will rewrite $p_1(n)$ replacing the contour γ by γ' in the integration defining p_1 , namely

$$(29) \quad p_1(n) = \int_{\gamma'} \sqrt{z/i} e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}} e^{-2\pi i n z} dz.$$

Next we simplify the triad of exponentials appearing in (29) by making a change of variables $z \mapsto \mu z$ so that their combination takes the form

$$e^{Ai(\frac{1}{z}-z)}.$$

This can be achieved under the two conditions $A = 2\pi\mu(n - \frac{1}{24})$ and $A = \frac{\pi}{12\mu}$, which means that

$$A = \frac{\pi}{\sqrt{6}}(n - \frac{1}{24})^{1/2} \quad \text{and} \quad \mu = \frac{1}{2\sqrt{6}}(n - \frac{1}{24})^{-1/2}.$$

Making the indicated change of variables we now have

$$(30) \quad p_1(n) = \mu^{3/2} \int_{\Gamma} e^{-sF(z)} \sqrt{z/i} \, dz,$$

with $F(z) = i(z - 1/z)$, $s = \frac{\pi}{\sqrt{6}}(n - \frac{1}{24})^{1/2}$. The curve Γ (see Figure 6). is now the union of three segments $[-a_n, -a_n + i\delta']$, $[-a_n + i\delta', a_n + i\delta']$, and $[a_n + i\delta', a_n]$; we can write $\Gamma = \mu^{-1}\gamma'$.

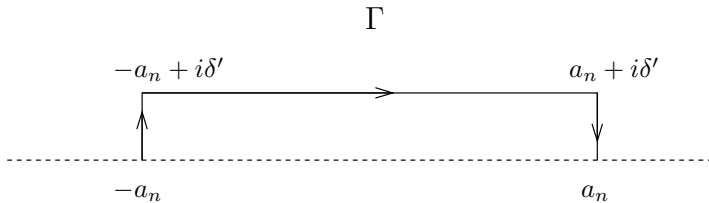


Figure 6. The curve Γ

Here $a_n = \frac{1}{2}\mu^{-1} = \sqrt{6}(n - \frac{1}{24})^{1/2} \approx n^{1/2}$, while $\delta' = \delta\mu^{-1} = \frac{2\sqrt{6}}{4\sqrt{6}n^{1/2}}(n - \frac{1}{24})^{1/2} \sim 1/2$, as $n \rightarrow \infty$.

We apply the method of steepest descent to the integral (30). In doing this, we note that $F(z) = i(z - 1/z)$ has one (complex) critical point $z = i$, in the upper half-plane. Moreover, the two curves passing through i on which F is real are: the imaginary axis, on which F has a maximum at $z = i$, which we reject, and the unit circle, on which F has a minimum at $z = i$. Thus using Cauchy's theorem we replace the integration on Γ by the integration over our final curve Γ^* , which consists of the segment $[-a_n, -1]$, $[1, a_n]$, together with the upper semicircle joining -1 to 1 .

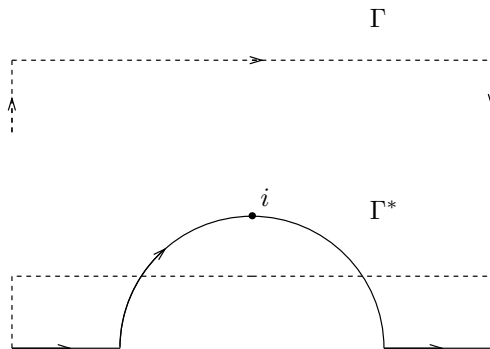


Figure 7. The final curve Γ^*

We therefore have

$$p_1(n) = \mu^{3/2} \int_{\Gamma^*} e^{-sF(z)} \sqrt{z/i} \, dz.$$

The contributions on the segments $[-a_n, -1]$ and $[1, a_n]$ are relatively very small, because on the real axis the exponential has absolute value 1, and hence the integrand is bounded by $\sup_{|z| \leq a_n} |z|^{1/2}$, and this leads to two terms which are $O(a_n^{3/2} \mu^{3/2}) = O(1)$.

Finally, we come to the principal part, which is the integration over the semicircle, taken with the orientation on the figure. Here we write $z = e^{i\theta}$, $dz = ie^{i\theta} d\theta$. Since $i(z - 1/z) = -2 \sin \theta$, this gives a contribution

$$-\mu^{3/2} \int_0^\pi e^{2s \sin \theta} e^{i3\theta/2} \sqrt{i} \, d\theta = \mu^{3/2} \int_{-\pi/2}^{\pi/2} e^{2s \cos \theta} (\cos(3\theta/2) + i \sin(3\theta/2)) d\theta.$$

In applying Proposition 2.1, Laplace's method, we take $\Phi(\theta) = -\cos \theta$, $\theta_0 = 0$, so $\Phi(\theta_0) = -1$, $\Phi''(\theta_0) = 1$ and we choose $\psi(\theta) = \cos(3\theta/2) + i \sin(3\theta/2)$, so that $\psi(\theta_0) = 1$. Therefore, the above contributes

$$\mu^{3/2} e^{2s} \frac{\sqrt{2\pi}}{(2s)^{1/2}} (1 + O(s^{-1/2})).$$

Now since $s = \frac{\pi}{\sqrt{6}} (n - \frac{1}{24})^{1/2}$, $\frac{2\pi}{\sqrt{6}} = \pi \sqrt{\frac{2}{3}} = K$, and $\mu = \frac{\sqrt{6}}{12} (n - \frac{1}{24})^{-1/2}$, we obtain

$$p(n) = \frac{1}{4n\sqrt{3}} e^{K n^{1/2}} (1 + O(n^{-1/4})),$$

and the first conclusion of the theorem is established.

To obtain the more exact conclusion (ii), we retrace our steps and use an additional device, which allows us to evaluate rather precisely the key integral. With $p_1(n)$ defined by (29), which is an integral taken over $\gamma' = \gamma'_n$, we write

$$p_1(n) = \frac{d}{dn}q(n) + e(n),$$

where

$$q(n) = \frac{1}{2\pi} \int_{\gamma'} (z/i)^{-1/2} e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}} e^{-2\pi inz} dz,$$

and $e(n)$ is the term due to the variation of the contour $\gamma' = \gamma'_n$, when forming the derivative in n . By Cauchy's theorem this is easily seen to be dominated by $O(e^{2\pi n\delta})$, which we have seen is $O(e^{\frac{K}{4}n^{1/2}})$, and can be subsumed in the error term. To analyze $q(n)$, we proceed as before, first making the change of variables $z \mapsto \mu z$, and then replacing the resulting contour Γ by Γ^* . As a consequence, we have

$$(31) \quad q(n) = \frac{\mu^{1/2}}{2\pi} \int_{\Gamma^*} e^{-sF(z)} (z/i)^{-1/2} dz,$$

with $F(z) = i(z - 1/z)$, $s = \frac{\pi}{\sqrt{6}}(n - \frac{1}{24})^{1/2}$, and $\mu = \frac{1}{2\sqrt{6}}(n - \frac{1}{24})^{-1/2}$.

Now the two segments $[-a_n, -1]$ and $[1, a_n]$ of the contour Γ^* make harmless contributions to $\frac{d}{dn}q(n)$, since F is purely imaginary on the real axis. Indeed, they yield terms which are $O(a_n^{1/2}\mu^{1/2}) = O(1)$.

The main part of (31) is the term arising from the integration on the semicircle. Thus setting $z = e^{i\theta}$, $dz = ie^{i\theta} d\theta$, and $i(z - 1/z) = -2\sin\theta$, it equals

$$\begin{aligned} -\frac{\mu^{1/2}}{2\pi} \int_0^\pi e^{2s\sin\theta} e^{i\theta/2} i^{3/2} d\theta &= \frac{\mu^{1/2}}{2\pi} \int_{-\pi/2}^{\pi/2} e^{2s\cos\theta} (\cos(\theta/2) + i\sin(\theta/2)) d\theta \\ &= \frac{\mu^{1/2}}{2\pi} \int_{-\pi/2}^{\pi/2} e^{2s\cos\theta} \cos(\theta/2) d\theta, \end{aligned}$$

where we have used the fact that the integral $\int_{-\pi/2}^{\pi/2} e^{2s\cos\theta} \sin(\theta/2) d\theta$ vanishes since the integrand is odd.

Now $\cos \theta = 1 - 2(\sin \theta/2)^2$, so setting $x = \sin(\theta/2)$ we see that the above integral becomes

$$\frac{\mu^{1/2} e^{2s}}{\pi} \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} e^{-4sx^2} dx.$$

However

$$\begin{aligned} \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} e^{-4sx^2} dx &= \int_{-\infty}^{\infty} e^{-4sx^2} dx + O\left(\int_{\frac{\sqrt{2}}{2}}^{\infty} e^{-4sx^2} dx\right) \\ &= \frac{\sqrt{\pi}}{2s^{1/2}} + O(e^{-2s}), \end{aligned}$$

and also

$$\frac{d}{ds} \left(\int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} e^{-4sx^2} dx \right) = \frac{d}{ds} \left(\frac{\sqrt{\pi}}{2s^{1/2}} \right) + O(e^{-2s}).$$

Gathering all the error terms together, we find

$$p(n) = \frac{d}{dn} \left(\mu^{1/2} \frac{e^{2s}}{\pi} \frac{\sqrt{\pi}}{2s^{1/2}} \right) + O(e^{\frac{K}{2}n^{1/2}}).$$

Since $s = \frac{\pi}{\sqrt{6}}(n - \frac{1}{24})^{1/2}$, $\mu = \frac{\sqrt{6}}{12}(n - \frac{1}{24})^{-1/2}$, and $K = \pi\sqrt{\frac{2}{3}}$, this is

$$p(n) = \frac{1}{2\pi\sqrt{2}} \frac{d}{dn} \left(\frac{e^{K(n-\frac{1}{24})^{1/2}}}{(n - \frac{1}{24})^{1/2}} \right) + O(e^{\frac{K}{2}n^{1/2}}),$$

and the theorem is proved.

5 Problems

1. Let η be an indefinitely differentiable function supported in a finite interval, so that $\eta(x) = 1$ for x near 0. Then, if $m > -1$ and $N > 0$,

$$\int_0^\infty e^{-sx} x^m \eta(x) dx = s^{-m-1} \Gamma(m+1) + O(s^{-N})$$

for $\operatorname{Re}(s) \geq 0$, $|s| \rightarrow \infty$.

(a) Consider first the case $-1 < m \leq 0$. It suffices to see that

$$\int_0^\infty e^{-sx} x^m (1 - \eta(x)) dx = O(s^{-N}),$$

and this can be done by repeated integration by parts since $e^{-sx} = (-1)^N s^{-N} \left(\frac{d}{dx}\right)^N (e^{-sx})$.

(b) To extend this to all m , find an integer k so that $k - 1 < m \leq k$, write

$$\int \left[\left(\frac{d}{dx} \right)^k (x^m) \right] e^{-sx} \eta(x) dx = c_{k,m} s^{-m+k-1} + O(s^{-N}),$$

and integrate by parts k times.

2. The following is a more precise version of Stirling's formula. There are real constant $a_1 = 1/12$, $a_2, \dots, a_n, \dots$, so that for every $N > 0$

$$\Gamma(s) = e^{s \log s} e^{-s} \frac{\sqrt{2\pi}}{s^{1/2}} \left(1 + \sum_{j=1}^N a_j s^{-j} + O(s^{-N}) \right) \quad \text{when } s \in S_\delta.$$

This can be proved by using the results of Problem 1 in place of Proposition 1.3.

3. The Bessel functions and Airy function have the following power series expansions:

$$J_\nu(x) = \left(\frac{x}{2} \right)^\nu \sum_{m=0}^{\infty} \frac{(-1)^m \left(\frac{x^2}{4} \right)^m}{m! \Gamma(\nu + m + 1)},$$

$$\text{Ai}(-x) = \frac{1}{\pi} \sum_{n=0}^{\infty} \frac{x^n}{n!} \sin(2\pi(n+1)/3) 3^{n/3-2/3} \Gamma(n/3 + 1/3).$$

(a) From this, verify that when $x > 0$,

$$\text{Ai}(-x) = \frac{x^{1/2}}{3} \left(J_{1/3} \left(\frac{2}{3} x^{3/2} \right) + J_{-1/3} \left(\frac{2}{3} x^{3/2} \right) \right).$$

(b) The function $\text{Ai}(x)$ extends to an entire function of order $3/2$.

[Hint: For (b), use (a), or alternatively, apply Problem 4 in Chapter 5 to the power series for Ai . Compare also with Problem 1, Chapter 4.]

4. Suppose $F(z) = \sum_{n=0}^{\infty} F_n w^n$ is meromorphic in a region containing the closed unit disc, and the only poles of F are on the unit circle at the points $\alpha_1, \dots, \alpha_k$, and their orders are $r_1, \dots, r_k$ respectively. Then for some $\epsilon > 0$

$$F_n = \sum_{j=1}^k P_j(n) + O(e^{-\epsilon n}) \quad \text{as } n \rightarrow \infty.$$

Here

$$P_j(n) = \frac{1}{(r_j - 1)!} \left(\frac{d}{dw} \right)^{r_j - 1} [(w - \alpha_j)^{r_j} w^{-n-1} F(w)]_{w=\alpha_j}.$$

Note that each P_j is of the form $P_j(n) = A_j(\alpha_j^{-1}n)^{r_j-1} + O(n^{r_j-2})$.

To prove this, use the residue formula (Theorem 1.4, Chapter 3).

5.* The one shortcoming in our derivation of the asymptotic formula for $p(n)$ arose from the fact that while $f_1(z) = \sqrt{z/i} e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}}$ is a good approximation to the generating function $f(z)$ near $z = 0$, this fails near other points on the real axis, since f_1 is regular there, but f is not.

However, using the transformation law (24) and the identity $f(z+1) = f(z)$, one can derive the following generalization of (24): whenever p/q is a rational number in lowest form (so p and q are relatively prime) then

$$f\left(z - \frac{p}{q}\right) = \omega_{p/q} \sqrt{\frac{zq}{i}} e^{\frac{i\pi}{12zq^2}} e^{-\frac{i\pi z}{12}} f\left(-\frac{1}{zq^2} - \frac{p'}{q}\right),$$

where $pp' = 1 \pmod{q}$. Here $\omega_{p/q}$ is an appropriate 24^{th} root of unity. This formula leads to an analogous $f_{p/q}$, approximating f at $z = p/q$.

From this one can obtain for each p/q a contribution of the form

$$c_{p/q} \frac{1}{2\pi\sqrt{2}} \frac{d}{dn} \left(\frac{e^{\frac{K}{q}(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}} \right)$$

to the asymptotic formula for $p(n)$. When suitably modified, the resulting series, summed over all proper fractions p/q in $[0, 1)$, actually converges and gives an exact formula for $p(n)$.

Appendix B: Simple Connectivity and Jordan Curve Theorem

Jordan was one of the precursors of the theory of functions of a real variable. He introduced in this part of analysis the capital notion of functions of bounded variation. Not less celebrated is his study of curves, universally called Jordan curves, which curves separate the plane in two distinct regions. We also owe him important propositions regarding the measure of sets that have led the way to numerous modern researches.

E. Picard, 1922

The notion of simple connectivity is at the source of many basic and fundamental results in complex analysis. To clarify the meaning of this important concept, we have gathered in this appendix some further insights into the properties of simply connected sets. Closely tied to the idea of simple connectivity is the notion of the “interior” of a simple closed curve. The theorem of Jordan states that this interior is well-defined and is simply connected. We prove here the special case of this theorem for curves which are piecewise-smooth.

Recall the definition in Chapter 3, according to which a region Ω is simply connected if any two curves in Ω with the same end-points are homotopic. From this definition we deduced an important version of Cauchy’s theorem which states that if Ω is simply connected and $\gamma \subset \Omega$ is any closed curve, then

$$(1) \quad \int_{\gamma} f(\zeta) d\zeta = 0$$

whenever f is holomorphic in Ω . Here, we shall prove that a converse also holds, therefore:

- (I) A region Ω is simply connected if and only if it is holomorphically simply connected; that is, whenever $\gamma \subset \Omega$ is closed and f holomorphic in Ω then (1) holds.

Besides this fundamental equivalence, which is analytic in nature, there

are also topological conditions that can be used to describe simple connectivity. In fact, the definition in terms of homotopies suggests that a simply connected set has no “holes.” In other words, one cannot find a closed curve in Ω that loops around points that do not belong to Ω . In the first part of this appendix we shall also turn these intuitive statements into tangible theorems:

- (II) We show that a bounded region Ω is simply connected if and only if its complement is connected.
- (III) We define the winding number of a curve around a point, and prove that Ω is simply connected if and only if no curve in Ω winds around points in the complement of Ω .

In the second part of this appendix we return to the problem of curves and their interior. The main question is the following: given a closed curve Γ that does not intersect itself (it is simple), can we make sense of the “region enclosed by Γ ”? In other words, what is the “interior” of Γ ? Naturally, we may expect the interior to be open, bounded, simply connected, and have Γ as its boundary. To solve this problem, at least when the curve is piecewise-smooth, we prove a theorem that guarantees the existence of a unique set which satisfies all the desired properties. This is a special case of the Jordan curve theorem, which is valid in the general case when the simple curve is assumed to be merely continuous. In particular, our result leads to a generalization of Cauchy’s theorem in Chapter 2 which we formulated for toy contours.

We continue to follow the convention set in Chapter 1 by using the term “curve” synonymously with “piecewise-smooth curve,” unless stated otherwise.

1 Equivalent formulations of simple connectivity

We first dispose of (I).

Theorem 1.1 *A region Ω is holomorphically simply connected if and only if Ω is simply connected.*

Proof. One direction is simply the version of Cauchy’s theorem in Corollary 5.3, Chapter 3. Conversely, suppose that Ω is holomorphically simply connected. If $\Omega = \mathbb{C}$, then it is clearly simply connected. If Ω is not all of $\mathbb{C}$, recall that the Riemann mapping theorem still applies (see the remark following its proof in Chapter 8), hence Ω is conformally equivalent to the unit disc. Since the unit disc is simply connected, the same must be true of Ω .

Next, we turn to (II) and (III), which, as we mentioned, are both precise formulations of the fact that a simply connected region cannot have “holes.”

Theorem 1.2 *If Ω is a bounded region in $\mathbb{C}$, then Ω is simply connected if and only if the complement of Ω is connected.*

Note that we assume that Ω is bounded. If this is not the case, then the theorem as stated does not hold, for example an infinite strip is simply connected yet its complement consists of two components. However, if the complement is taken with respect to the extended complex plane, that is, the Riemann sphere, then the conclusion of the theorem holds regardless of whether Ω is bounded or not.

Proof. We begin with the proof that if Ω^c is connected, then Ω is simply connected. This will be achieved by showing that Ω is holomorphically simply connected. Therefore, let γ be a closed curve in Ω and f a holomorphic function on Ω . Since Ω is bounded, the set¹

$$K = \{z \in \Omega : d(z, \Omega^c) \geq \epsilon\}$$

is compact, and for sufficiently small ϵ , the set K contains γ . In an attempt to apply Runge’s theorem (Theorem 5.7 in Chapter 2), we must first show that the complement K^c of K is connected.

If this is not the case, then K^c is the disjoint union of two non-empty open sets, say $K^c = \mathcal{O}_1 \cup \mathcal{O}_2$. Let

$$F_1 = \mathcal{O}_1 \cap \Omega^c \quad \text{and} \quad F_2 = \mathcal{O}_2 \cap \Omega^c.$$

Clearly, $\Omega^c = F_1 \cup F_2$, so if we can show that F_1 and F_2 are disjoint, closed, and non-empty, then we will conclude that Ω^c is not connected, thus contradicting the hypothesis in the theorem. Since $\mathcal{O}_1$ and $\mathcal{O}_2$ are disjoint, so are F_1 and F_2 . To see why F_1 is closed, suppose $\{z_n\}$ is a sequence of points in F_1 that converges to z . Since Ω^c is closed we must have $z \in \Omega^c$, and since Ω^c is at a finite distance from K , we deduce that $z \in \mathcal{O}_1 \cup \mathcal{O}_2$. Now we observe that we cannot have $z \in \mathcal{O}_2$, for otherwise we would have $z_n \in \mathcal{O}_2$ for sufficiently large n because $\mathcal{O}_2$ is open, and this contradicts the fact that $z_n \in F_1$ and $\mathcal{O}_1 \cap \mathcal{O}_2 = \emptyset$. Hence $z \in \mathcal{O}_1$ and F_1 is closed, as desired. Finally, we claim that F_1 is non-empty. If otherwise, $\mathcal{O}_1$ is contained in Ω . Select any point $w \in \mathcal{O}_1$, and since $w \notin K$, there exists $z \in \Omega^c$ with $|w - z| < \epsilon$, and the entire line segment from w to z belongs to K^c . Since $z \in \mathcal{O}_2$ (because $\mathcal{O}_1 \subset \Omega$), some point

¹Here, $d(z, \Omega^c) = \inf\{|z - w| : w \in \Omega^c\}$ denotes the distance from z to Ω^c .

on the line segment $[z, w]$ must belong to neither $\mathcal{O}_1$ nor $\mathcal{O}_2$, and this is a contradiction. More precisely, if we set

$$t^* = \sup\{0 \leq t \leq 1 : (1-t)z + tw \in \mathcal{O}_2\},$$

then $0 < t^* < 1$, and the point $(1-t^*)z + t^*w$, which is not in K , cannot belong to either $\mathcal{O}_1$ or $\mathcal{O}_2$ since these sets are open. Similar arguments imply the same conclusions for F_2 , and we have reached the desired contradiction. Thus K^c is connected.

Therefore, Runge's theorem guarantees that f can be approximated uniformly on K , and hence on γ , by polynomials. However, $\int_\gamma P(z) dz = 0$ whenever P is a polynomial, so in the limit we conclude that $\int_\gamma f(z) dz = 0$, as desired.

The converse result, that Ω^c is connected whenever Ω is bounded and simply connected, will follow from the notion of winding numbers, which we discuss next.

Winding numbers

If γ is a closed curve in $\mathbb{C}$ and z a point not lying on γ , then we may calculate the number of times the curve γ winds around z by looking at the change of argument of the quantity $\zeta - z$ as ζ travels on γ . Every time γ loops around z , the quantity $(1/2\pi) \arg(\zeta - z)$ increases (or decreases) by 1. If we recall that $\log w = \log |w| + i \arg w$, and denote the beginning and ending points of γ by ζ_1 and ζ_2 , then we may guess that the quantity

$$\frac{1}{2\pi i} [\log(\zeta_1 - z) - \log(\zeta_2 - z)], \quad \text{which "equals"} \quad \frac{1}{2\pi i} \int_\gamma \frac{d\zeta}{\zeta - z},$$

computes precisely the number of times γ loops around ζ .

These considerations lead to the following precise definition: the **winding number** of a closed curve γ around a point $z \notin \gamma$ is

$$W_\gamma(z) = \frac{1}{2\pi i} \int_\gamma \frac{d\zeta}{\zeta - z}.$$

Sometimes, $W_\gamma(z)$ is also called the index of z with respect to γ .

For example, if $\gamma(t) = e^{ikt}$, $0 \leq t \leq 2\pi$, is the unit circle traversed k times in the positive direction (with $k \in \mathbb{N}$), then $W_\gamma(0) = k$. In fact, one has

$$W_\gamma(z) = \begin{cases} k & \text{if } |z| < 1, \\ 0 & \text{if } |z| > 1. \end{cases}$$

Similarly, if $\gamma(t) = e^{-ikt}$, $0 \leq t \leq 2\pi$, is the unit circle traversed k times in the negative direction, then we find that $W_\gamma(z) = -k$ in the interior of the disc, and $W_\gamma(z) = 0$ in its exterior.

Note that, if γ denotes a positively oriented toy contour, then

$$W_\gamma(z) = \begin{cases} 1 & \text{if } z \in \text{interior of } \gamma, \\ 0 & \text{if } z \in \text{exterior of } \gamma. \end{cases}$$

In general we have the following natural facts about winding numbers.

Lemma 1.3 *Let γ be a closed curve in $\mathbb{C}$.*

- (i) *If $z \notin \gamma$, then $W_\gamma(z) \in \mathbb{Z}$.*
- (ii) *If z and w belong to the same open connected component in the complement of γ , then $W_\gamma(z) = W_\gamma(w)$.*
- (iii) *If z belongs to the unbounded connected component in the complement of γ , then $W_\gamma(z) = 0$.*

Proof. To see why (i) is true, suppose that $\gamma : [0, 1] \rightarrow \mathbb{C}$ is a parametrization for the curve, and let

$$G(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s) - z} ds.$$

Then G is continuous and, except possibly at finitely many points, it is differentiable with $G'(t) = \gamma'(t)/(\gamma(t) - z)$. This implies that, except possibly at finitely many points, the derivative of the continuous function $H(t) = (\gamma(t) - z)e^{-G(t)}$ is zero, and hence H must be constant. Putting $t = 0$ and recalling that γ is closed, so that $\gamma(0) = \gamma(1)$, we find

$$1 = e^{G(0)} = c(\gamma(0) - z) = c(\gamma(1) - z) = e^{G(1)}.$$

Therefore, $G(1)$ is an integral multiple of $2\pi i$, as desired.

For (ii), we simply note that $W_\gamma(z)$ is a continuous function of $z \notin \gamma$ that is integer-valued, so it must be constant in any open connected component in the complement of γ .

Finally, one observes that $\lim_{|z| \rightarrow \infty} W_\gamma(z) = 0$, and, combined with (ii), this establishes (iii).

We now show that the notion of a bounded simply connected set Ω may be understood in the following sense: no curve in Ω winds around points in Ω^c .

Theorem 1.4 *A bounded region Ω is simply connected if and only if $W_\gamma(z) = 0$ for any closed curve γ in Ω and any point z not in Ω .*

Proof. If Ω is simply connected and $z \notin \Omega$, then $f(\zeta) = 1/(\zeta - z)$ is holomorphic in Ω , and Cauchy's theorem gives $W_\gamma(z) = 0$.

For the converse, it suffices to prove that the complement of Ω is connected (Theorem 1.2). We argue by contradiction, and construct an explicit closed curve γ in Ω and find a point w so that $W_\gamma(w) \neq 0$.

If we suppose that Ω^c is not connected, then we may write $\Omega^c = F_1 \cup F_2$ where F_1, F_2 are disjoint, closed, and non-empty. Only one of these sets can be unbounded, so that we may assume that F_1 is bounded, thus compact. The curve γ will be constructed as part of the boundary of an appropriate union of squares.

Lemma 1.5 *Let w be any point in F_1 . Under the above assumptions, there exists a finite collection of closed squares $\mathcal{Q} = \{Q_1, \dots, Q_n\}$ that belong to a uniform grid $\mathcal{G}$ of the plane, and are such that:*

- (i) w belongs to the interior of Q_1 .
- (ii) The interiors of Q_j and Q_k are disjoint when $j \neq k$.
- (iii) F_1 is contained in the interior of $\bigcup_{j=1}^n Q_j$.
- (iv) $\bigcup_{j=1}^n Q_j$ is disjoint from F_2 .
- (v) The boundary of $\bigcup_{j=1}^n Q_j$ lies entirely in Ω , and consists of a finite union of disjoint simple closed polygonal curves.

Assuming this lemma for now, we may easily finish the proof of the theorem. The boundary ∂Q_j of each square is equipped with the positive orientation. Since $w \in Q_1$, and $w \notin Q_j$ for all $j > 1$, we have

$$(2) \quad \sum_{j=1}^n \frac{1}{2\pi i} \int_{\partial Q_j} \frac{d\zeta}{\zeta - w} = 1.$$

If $\gamma_1, \dots, \gamma_M$ denotes the polygonal curves in (v) of the lemma, then, the cancellations arising from integrating over the same side but in opposite directions in (2) yield

$$\sum_{j=1}^n \frac{1}{2\pi i} \int_{\gamma_j} \frac{d\zeta}{\zeta - w} = 1,$$

and hence $W_{\gamma_{j_0}}(w) \neq 0$ for some j_0 . The closed curve γ_{j_0} lies entirely in Ω , and this gives the desired contradiction.

Proof of the lemma. Since F_2 is closed, the sets F_1 and F_2 are at a finite non-zero distance d from one another. Now consider a uniform grid $\mathcal{G}_0$ of the plane consisting of closed squares of side length which is much smaller than d , say $< d/100$, and such that w lies at the center of a closed square R_1 in this grid. Let $\mathcal{R} = \{R_1, \dots, R_m\}$ denote the finite collection of all closed squares in the grid that intersect F_1 . Then, the collection $\mathcal{R}$ satisfies properties (i) through (iv) of the lemma. To guarantee (v), we argue as follows.

The boundary of each square in $\mathcal{R}$ is given the positive (counterclockwise) orientation. The boundary of $\bigcup_{j=1}^m R_j$ is then equal to the union of all boundary sides, that is, those sides that do not belong to two adjacent squares in the collection $\mathcal{R}$. Similarly, the boundary vertices are the end-points of all boundary sides. A boundary vertex is said to be “bad,” if it is the end-point of more than two boundary sides. (See point P on Figure 1.)

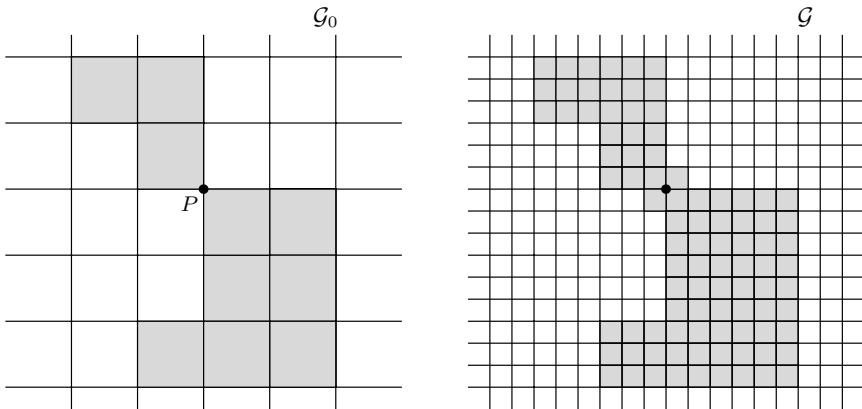


Figure 1. Eliminating bad boundary vertices

To eliminate the bad boundary vertices, we refine the grid $\mathcal{G}_0$ and possibly add some squares. More precisely, consider the grid $\mathcal{G}$ obtained as a refinement of the original grid, by dissecting all squares of $\mathcal{G}_0$ into nine equal subsquares. Then, let $Q_1, \dots, Q_p$ denote all the squares in the grid $\mathcal{G}$ that are subsquares of squares in the collection $\mathcal{R}$ (so in particular, $p = 9n$), and where Q_1 is chosen so that $w \in Q_1$. Then, we may add finitely many squares from $\mathcal{G}$ near each bad boundary vertex, so that the resulting family $\mathcal{Q} = \{Q_1, \dots, Q_n\}$ has no bad boundary vertices. (See Figure 1.)

Clearly, $\mathcal{Q}$ still satisfies (i) through (iv), and we claim this collection also satisfies (v). Indeed, let $[a_1, a_2]$ denote any boundary side of $\bigcup_{j=1}^n Q_j$ with its orientation from a_1 to a_2 . By considering the three different possibilities, one sees that a_2 is the beginning point of another boundary side $[a_2, a_3]$. Continuing in this fashion, we obtain a sequence of boundary sides $[a_1, a_2], [a_2, a_3], \dots, [a_n, a_{n+1}], \dots$. Since there are only finitely many sides, we must have $a_n = a_m$ for some n and some $m > n$. We may choose the smallest m so that $a_n = a_m$, say $m = m'$. Then, we note that if $n > 1$, then $a_{m'}$ is an end-point of at least three boundary sides, namely $[a_{n-1}, a_n]$, $[a_n, a_{n+1}]$, and $[a_{m'-1}, a_{m'}]$, hence $a_{m'}$ is a bad boundary vertex. Since we arranged that $\mathcal{Q}$ had no such boundary vertices, we conclude that $n = 1$, and hence the polygon formed by $a_1, \dots, a_{m'}$ is closed and simple. We may repeat this process and find that $\mathcal{Q}$ satisfies property (v), and the proof of Lemma 1.5 is complete.

Finally, we are now able to finish the proof of Theorem 1.2, namely, if Ω is bounded and simply connected we can conclude that Ω^c is connected. To see this, note that if Ω^c is not connected, then we have constructed a curve $\gamma \subset \Omega$ and found a point $w \notin \Omega$ so that $W_\gamma(w) \neq 0$, thus contradicting the fact that Ω is simply connected.

2 The Jordan curve theorem

Although we emphasize in the statement of the theorems which follow that the curves are piecewise-smooth, we note that the proofs involve the use of curves that may only be continuous, (the curves Γ_ϵ below).

The two main results in this section are the following.

Theorem 2.1 *Let Γ be curve in the plane that is simple and piecewise-smooth. Then, the complement of Γ is an open connected set whose boundary is precisely Γ .*

Theorem 2.2 *Let Γ be a curve in the plane which is simple, closed, and piecewise-smooth. Then, the complement of Γ consists of two disjoint connected open sets. Precisely one of these regions is bounded and simply connected; it is called the **interior** of Γ and denoted by Ω . The other component is unbounded, called the **exterior** of Γ , and denoted by $\mathcal{U}$.*

Moreover, with the appropriate orientation for Γ , we have

$$W_\Gamma(z) = \begin{cases} 1 & \text{if } z \in \Omega, \\ 0 & \text{if } z \in \mathcal{U}. \end{cases}$$

Remark. These two theorems continue to hold in the general case where we drop the assumption that the curves are piecewise-smooth. However, as it turns out, the proofs then are more difficult. Fortunately, the restricted setting of piecewise-smooth curves suffices for many applications.

As a consequence of the above propositions, we may state a version of Cauchy's theorem as follows:

Theorem 2.3 *Suppose f is a function that is holomorphic in the interior Ω of a simple closed curve Γ . Then*

$$\int_{\eta} f(\zeta) d\zeta = 0$$

whenever η is any closed curve contained in Ω .

The idea of the proof of Theorem 2.1 can be roughly summarized as follows. Since the complement of Γ is open, it is sufficient to show it is pathwise connected (Exercise 5, Chapter 1). Let z and w belong to the complement of Γ , and join these two points by a curve. If this curve intersects Γ , we first connect z to z' and w to w' , where z' and w' are close to Γ , by curves that do not intersect Γ . Then, we join z' to w' by traveling “parallel” to the curve Γ and going around its end-points if necessary.

Therefore, the key is to construct a family of continuous curves that are “parallel” to Γ . This can be achieved because of the conditions imposed on the curve. Indeed, if γ is a parametrization for a smooth piece of Γ , then γ is continuously differentiable, and $\gamma'(t) \neq 0$. Moreover, the vector $\gamma'(t)$ is tangent to Γ . Consequently, $i\gamma'(t)$ is perpendicular to Γ , and if Γ is simple, considering $\gamma(t) + i\epsilon\gamma'(t)$ amounts to a new curve that is “parallel” to Γ . The details are as follows.

In the next three lemmas and two propositions, we emphasize that Γ_0 denotes a simple *smooth* curve. We recall that an arc-length parametrization γ for a smooth curve Γ_0 satisfies $|\gamma'(t)| = 1$ for all t . Every smooth curve has an arc-length parametrization.

Lemma 2.4 *Let Γ_0 be a simple smooth curve with an arc-length parametrization given by $\gamma : [0, L] \rightarrow \mathbb{C}$. For each real number ϵ , let Γ_ϵ be the continuous curve defined by the parametrization*

$$\gamma_\epsilon(t) = \gamma(t) + i\epsilon\gamma'(t), \quad \text{for } 0 \leq t \leq L.$$

Then, there exists $\kappa_1 > 0$ so that $\Gamma_0 \cap \Gamma_\epsilon = \emptyset$ whenever $0 < |\epsilon| < \kappa_1$.

Proof. We first prove the result locally. If s and t belong to $[0, L]$, then

$$\begin{aligned}\gamma_\epsilon(t) - \gamma(s) &= \gamma(t) - \gamma(s) + i\epsilon\gamma'(t) \\ &= \int_s^t \gamma'(u) du + i\epsilon\gamma'(t) \\ &= \int_s^t [\gamma'(u) - \gamma'(t)] du + (t - s + i\epsilon)\gamma'(t).\end{aligned}$$

Since γ' is uniformly continuous on $[0, L]$, there exists $\delta > 0$ so that $|\gamma'(x) - \gamma'(y)| < 1/2$ whenever $|x - y| < \delta$. In particular, if $|s - t| < \delta$ we find that

$$|\gamma_\epsilon(t) - \gamma(s)| > |t - s + i\epsilon| |\gamma'(t)| - \frac{|t - s|}{2}.$$

Since γ is an arc-length parametrization, we have $|\gamma'(t)| = 1$, and hence

$$|\gamma_\epsilon(t) - \gamma(s)| > |\epsilon|/2,$$

where we have used the simple fact that $2|a + ib| \geq |a| + |b|$ whenever a and b are real. This proves that $\gamma_\epsilon(t) \neq \gamma(s)$ whenever $|t - s| < \delta$ and $\epsilon \neq 0$.

To conclude the proof of the lemma, we argue as follows. (See Figure 2 for an illustration of the argument.)

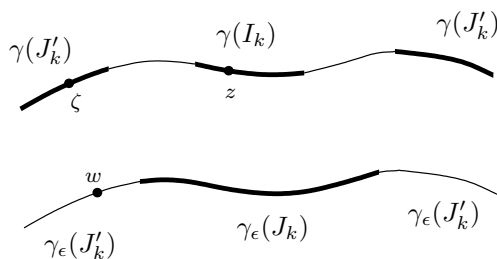


Figure 2. Situation in the proof of Lemma 2.4

Let $0 = t_0 < \dots < t_n = L$ be a partition of $[0, L]$ with $|t_{k+1} - t_k| < \delta$ for all k , and consider

$$I_k = \{t : |t - t_k| \leq \delta/4\}, \quad J_k = \{t : |t - t_k| \leq \delta/2\},$$

and

$$J'_k = \{t : |t - t_k| \geq \delta/2\}.$$

Then, we have just proved that

$$(3) \quad \gamma(I_k) \cap \gamma_\epsilon(J_k) = \emptyset \quad \text{whenever } \epsilon \neq 0.$$

Since Γ_0 is simple, the distance d_k between the two compact sets $\gamma(I_k)$ and $\gamma(J'_k)$ is strictly positive. We now claim that

$$(4) \quad \gamma(I_k) \cap \gamma_\epsilon(J'_k) = \emptyset \quad \text{whenever } |\epsilon| < d_k/2.$$

Indeed, if $z \in \gamma(I_k)$ and $w \in \gamma_\epsilon(J'_k)$, then we choose s in J'_k so that $w = \gamma_\epsilon(s)$ and let $\zeta = \gamma(s)$. The triangle inequality then implies

$$|z - w| \geq |z - \zeta| - |\zeta - w| \geq d_k - |\epsilon| \geq d_k/2,$$

and the claim is established. Finally, if we choose $\kappa_1 = \min_k d_k/2$, then (3) and (4) imply that $\Gamma_0 \cap \Gamma_\epsilon = \emptyset$ whenever $0 < |\epsilon| < \kappa_1$, as desired.

The next lemma shows that any point close to an interior point of the curve belongs to one of its parallel translates. By an interior point of the curve, we mean a point of the form $\gamma(t)$ with t in the open interval $(0, L)$. Such a point should not to be confused with an “interior” point of a curve, as in Theorem 2.2.

Lemma 2.5 *Suppose z is a point which does not belong to the smooth curve Γ_0 , but that is closer to an interior point of the curve than to either of its end-points. Then z belongs to Γ_ϵ for some $\epsilon \neq 0$.*

More precisely, if $z_0 \in \Gamma_0$ is closest to z and $z_0 = \gamma(t_0)$ for some t_0 in the open interval $(0, L)$, then $z = \gamma(t_0) + i\epsilon\gamma'(t_0)$ for some $\epsilon \neq 0$.

Proof. For t in a neighborhood of t_0 the fact that γ is differentiable guarantees that

$$z - \gamma(t) = z - \gamma(t_0) - \gamma'(t_0)(t - t_0) + o(|t - t_0|).$$

Since $z_0 = \gamma(t_0)$ minimizes the distance from z to Γ_0 , we find that

$$|z - z_0|^2 \leq |z - \gamma(t)|^2 = |z - z_0|^2 - 2(t - t_0)\operatorname{Re}\left([z - \gamma(t_0)]\overline{\gamma'(t_0)}\right) + o(|t - t_0|).$$

Since $t - t_0$ can take on positive or negative values, we must have $\operatorname{Re}\left([z - \gamma(t_0)]\overline{\gamma'(t_0)}\right) = 0$, otherwise the above inequality can be violated for t close to t_0 . As a result, there exists a real number ϵ with $[z - \gamma(t_0)]\overline{\gamma'(t_0)} = i\epsilon$. Since $|\gamma'(t_0)| = 1$ we have $\overline{\gamma'(t_0)} = 1/\gamma'(t_0)$, and therefore $z - \gamma(t_0) = i\epsilon\gamma'(t_0)$. The proof of the lemma is complete.

Suppose that z and w are close to interior points of Γ_0 , so that $z \in \Gamma_\epsilon$ and $w \in \Gamma_\eta$ for some non-zero ϵ and η . If ϵ and η have the same sign, we say that the points z and w belong to the same side of Γ_0 . Otherwise, z and w are said to be on opposite sides of Γ_0 . We stress the fact that we do not attempt to define the “two sides of Γ_0 ,” but only that given two points near Γ_0 , we may infer if they are on the “same side” or on “opposite sides”. Also, nothing we have done so far shows that these conditions are mutually exclusive.

Roughly speaking, points on the same side can be joined almost directly by a curve “parallel” to Γ_0 , while for points on opposite sides, we also need to go around one of the end-points of Γ_0 .

We first investigate the situation for points on the same side of Γ_0 .

Proposition 2.6 *Let A and B denote the two end-points of a simple smooth curve Γ_0 , and suppose that K is a compact set that satisfies either*

$$\Gamma_0 \cap K = \emptyset \quad \text{or} \quad \Gamma_0 \cap K = A \cup B.$$

If $z \notin \Gamma_0$ and $w \notin \Gamma_0$ lie on the same side of Γ_0 , and are closer to interior points of Γ_0 than they are to K or to the end-points of Γ_0 , then z and w can be joined by a continuous curve that lies entirely in the complement of $K \cup \Gamma_0$.

The unspecified compact set K will be chosen appropriately in the proof of the Jordan curve theorem.

Proof. By the previous lemma, consider $z_0 = \gamma(t_0)$ and $w_0 = \gamma(s_0)$ that are interior points of Γ_0 closest to z and w , respectively. Then

$$z = \gamma(t_0) + i\epsilon_0\gamma'(t_0) \quad \text{and} \quad w = \gamma(s_0) + i\eta_0\gamma'(s_0),$$

where ϵ_0 and η_0 have the same sign, which we may assume to be positive. We may also assume that $t_0 \leq s_0$.

The hypothesis of the lemma implies that the line segments joining z to z_0 and w to w_0 are entirely contained in the complement of K and Γ_0 . Therefore, for all small $\epsilon > 0$, we may join z and w to the points

$$z_\epsilon = \gamma(t_0) + i\epsilon\gamma'(t_0) \quad \text{and} \quad w = \gamma(s_0) + i\epsilon\gamma'(s_0),$$

respectively. See Figure 3.

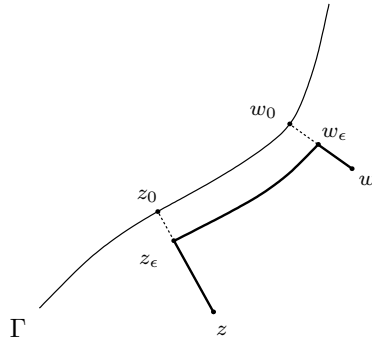


Figure 3. Situation in the proof of Proposition 2.6

Finally, if ϵ is chosen smaller than κ_1 in Lemma 2.4 and also smaller than the distance from K to the part of Γ_0 between z_0 and w_0 , that is, $\{\gamma(t) : t_0 \leq t \leq s_0\}$, then the corresponding part of Γ_ϵ , namely $\{\gamma_\epsilon(t) : t_0 \leq t \leq s_0\}$, joins the point z_ϵ to w_ϵ . Moreover, this curve is contained in the complement of K and Γ_0 . This proves the proposition.

To join points on opposite sides of Γ_0 , we need the following preliminary result, which ensures that there is enough room necessary to travel around the end-points.

Lemma 2.7 *Let Γ_0 be a simple smooth curve. There exists $\kappa_2 > 0$ so that the set N , which consists of points of the form $z = \gamma(L) + \epsilon e^{i\theta}\gamma'(L)$ with $-\pi/2 \leq \theta \leq \pi/2$ and $0 < \epsilon < \kappa_2$, is disjoint from Γ_0 .*

Proof. The argument is similar to the one given in the proof of Lemma 2.4. First, we note that

$$\gamma(L) + \epsilon e^{i\theta}\gamma'(L) - \gamma(t) = \int_t^L [\gamma'(u) - \gamma'(L)] du + (L - t + \epsilon e^{i\theta})\gamma'(L).$$

If we choose δ so that $|\gamma'(u) - \gamma'(L)| < 1/2$ when $|u - L| < \delta$, then $|t - L| < \delta$ implies

$$|\gamma(L) + \epsilon e^{i\theta} \gamma'(L) - \gamma(t)| \geq |\epsilon|/2.$$

Therefore $\gamma(t) \notin N$ whenever $L - \delta \leq t \leq L$. Finally, it suffices to choose κ_2 smaller than the distance from the end-point $\gamma(L)$ to the rest of the curve $\gamma(t)$ with $0 \leq t \leq L - \delta$, to conclude the proof.

Finally, we may state the result analogous to Proposition 2.6 for points that could lie on opposite sides of Γ_0 .

Proposition 2.8 *Let A denote an end-point of the simple smooth curve Γ_0 , and suppose that K is a compact set that satisfies either*

$$\Gamma_0 \cap K = \emptyset \quad \text{or} \quad \Gamma_0 \cap K = A.$$

If $z \notin \Gamma_0$ and $w \notin \Gamma_0$ are closer to interior points of Γ_0 than they are to K or to the end-points of Γ_0 , then z and w can be joined by a continuous curve that lies entirely in the complement of $\Gamma_0 \cup K$.

We only provide an outline of the argument, which is similar to the proof of Proposition 2.6. It suffices to consider the case when z and w lie on opposite sides of Γ_0 and $A = \gamma(0)$. First, we may join

$$z_\epsilon = \gamma(t_0) + i\epsilon\gamma'(t_0) \quad \text{and} \quad w_\epsilon = \gamma(s_0) - i\epsilon\gamma'(s_0)$$

to the points

$$z'_\epsilon = \gamma(L) + i\epsilon\gamma'(L) \quad \text{and} \quad w'_\epsilon = \gamma(L) - i\epsilon\gamma'(L).$$

Then, z'_ϵ and w'_ϵ may be joined within the “half-neighborhood” N of Lemma 2.7. Here, if $t_0 \leq s_0$ we must select $|\epsilon|$ smaller than the distance from $\{\gamma(t) : t_0 \leq t \leq L\}$ to K , and also smaller than κ_1 and κ_2 of Lemmas 2.4 and 2.7.

Proof of Theorem 2.1

Let Γ be a simple piecewise-smooth curve.

First, we prove that the boundary of the set $\mathcal{O} = \Gamma^c$ is precisely Γ . Clearly, $\mathcal{O}$ is an open set whose boundary is contained in Γ . Moreover, any point where Γ is smooth also belongs to the boundary of $\mathcal{O}$ (by Lemma 2.4 for instance). Since the boundary of $\mathcal{O}$ must also be closed, we conclude it is equal to all of Γ .

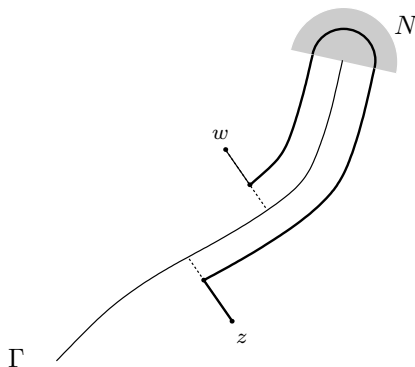


Figure 4. Situation in the proof of Proposition 2.8

The proof that $\mathcal{O}$ is connected is by induction on the number of smooth curves constituting Γ . Suppose first that Γ is simple and smooth, and let Z and W be any two points that do not lie on Γ . Let Λ be any smooth curve in $\mathbb{C}$ that joins Z and W , and which omits the two end-points of Γ . If Λ intersects Γ , it does so at interior points. Therefore, we may join Z by a piece of Λ that does not intersect Γ to a point z that is closer to the interior of Γ than to either of its end-points. Similarly, W can be joined in the complement of Γ to a point w also closer to the interior of Γ than to either of its end-points. Proposition 2.8 (with K empty) then shows that z and w can be joined by a continuous curve in the complement of Γ . Altogether, we may join any two points in the complement of Γ , and this proves the base step of the induction.

Suppose that the theorem is proved for all curves containing $n - 1$ smooth curves, and let Γ consist of n smooth curves, so that we may write

$$\Gamma = K \cup \Gamma_0,$$

where K is the union of $n - 1$ consecutive smooth curves, and Γ_0 is smooth. In particular, K is compact and intersects Γ_0 in a single one of its end-points. By the induction hypothesis, any two points Z and W in the complement of Γ can be joined by a curve that does not intersect K , and we may also assume that this curve omits both end-points of Γ_0 . If this curve intersects Γ_0 in its interior, then we apply Proposition 2.8 to conclude the proof of the theorem.

Proof of Theorem 2.2

Let Γ denote a curve which is simple, closed, and piecewise-smooth. We first prove that the complement of Γ consists of *at most* two components.

Fix a point W that lies outside some large disc that contains Γ , and let $\mathcal{U}$ denote the set of all points that can be joined to W by a continuous curve that lies entirely in the complement of Γ . The set $\mathcal{U}$ is clearly open, and also connected since any two points can be joined by passing first through W . Now we define

$$\Omega = \Gamma^c - \mathcal{U}.$$

We must show that Ω is connected. To this end, let K denote the curve obtained by deleting a smooth piece Γ_0 of Γ . By the Jordan arc theorem, we may join any point $Z \in \Omega$ to W by a curve Λ_Z that does not intersect K . Since $Z \notin \mathcal{U}$, the curve Λ_Z must intersect Γ_0 at one of its interior points. We may therefore choose two points $z, w \in \Lambda_Z$ closer to interior points of Γ_0 than to either of its end-points, and so that the pieces of Λ_Z joining Z to z and W to w are entirely contained in the complement of Γ . Then, the points z and w are on opposite sides of Γ_0 , for otherwise, we could apply Proposition 2.6 to find that Z can be joined to W by a curve lying in the complement of Γ , and this contradicts $Z \notin \mathcal{U}$. Finally, if Z_1 is another point in Ω , the two corresponding points z_1 and w_1 must also lie on opposite sides of Γ_0 . Moreover, z and z_1 must lie on the same side of Γ_0 , for otherwise z and w_1 do, and we can once again join Z to W without crossing Γ , thus contradicting $Z \notin \mathcal{U}$. Therefore, by Proposition 2.6 the points z and z_1 can be joined by a curve in the complement of Γ , and we conclude that Z and Z_1 belong to the same connected component.

The argument thus far proves that Γ^c contains at most two components, but nothing as yet guarantees that Ω is non-empty. To show that Γ^c has precisely two components, it suffices (by Lemma 1.3) to prove that there are points that have different winding numbers with respect to Γ . In fact, we claim that points that are on opposite sides of Γ have winding numbers that differ by 1. To see this, fix a point z_0 on a smooth part of Γ , say $z_0 = \gamma(t_0)$, let $\epsilon > 0$, and define

$$z_\epsilon = \gamma(t_0) + i\epsilon\gamma'(t_0) \quad \text{and} \quad w_\epsilon = \gamma(t_0) - i\epsilon\gamma'(t_0).$$

By our previous observations, points on the same side of Γ belong to the same connected component, and hence

$$\Delta = |W_\Gamma(z_\epsilon) - W_\Gamma(w_\epsilon)|$$

is constant for all small $\epsilon > 0$.

First, we may write

$$\left(\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} \right) = \frac{2i\epsilon\gamma'(t_0)\gamma'(t)}{[\gamma(t) - \gamma(t_0)]^2 + \epsilon^2\gamma'(t_0)^2}.$$

For the numerator, we use

$$\begin{aligned}\gamma'(t) &= \gamma'(t_0) + [\gamma'(t) - \gamma'(t_0)] \\ &= \gamma'(t_0) + \psi(t),\end{aligned}$$

where $\psi(t) \rightarrow 0$ as $t \rightarrow t_0$. For the denominator, we recall that $\gamma'(t_0) \neq 0$, so that

$$[\gamma(t) - \gamma(t_0)]^2 + \epsilon^2\gamma'(t_0)^2 = \gamma'(t_0)^2[(t - t_0)^2 + \epsilon^2] + o(|t - t_0|).$$

Putting these results together, we see that

$$\left(\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} \right) = \frac{2i\epsilon}{(t - t_0)^2 + \epsilon^2} + E(t),$$

where given $\eta > 0$, there exists $\delta > 0$ so that if $|t - t_0| \leq \delta$, the error term satisfies

$$|E(t)| \leq \eta \frac{\epsilon}{(t - t_0)^2 + \epsilon^2}.$$

We then write

$$\begin{aligned}\Delta &= \frac{1}{2\pi i} \int_{|t-t_0| \geq \delta} \left(\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} \right) dt + \\ &\quad + \frac{1}{2\pi i} \int_{|t-t_0| < \delta} \left(\frac{2i\epsilon}{(t - t_0)^2 + \epsilon^2} + E(t) \right) dt.\end{aligned}$$

The first integral goes to 0 as $\epsilon \rightarrow 0$. In the second integral we make the change of variables $t - t_0 = \epsilon s$, and note that

$$\frac{1}{\pi} \int_{-\rho}^{\rho} \frac{ds}{s^2 + 1} = \frac{1}{\pi} [\arctan s]_{-\rho}^{\rho} \rightarrow 1 \quad \text{as } \rho \rightarrow \infty.$$

We therefore see that letting $\epsilon \rightarrow 0$ gives

$$|\Delta - 1| < \eta.$$

We conclude that $\Delta = 1$, and hence Γ^c has precisely two components. Finally, only one of these components can be unbounded, namely $\mathcal{U}$, and the winding number of Γ in this component must therefore be zero. By our last result, we see that, after possibly reversing the orientation of the curve, the winding number of any point in the bounded component Ω is constant and equal to 1. Also, it is clear from what has been said that any smooth point on Γ can be approached by points in either component, and hence Γ is the boundary of both Ω and $\mathcal{U}$.

The final step in the proof is to show that the interior of the curve, that is, the bounded component Ω , is simply connected. By Theorem 1.2 it suffices to show that Ω^c is connected. If not, then

$$\Omega^c = F_1 \cup F_2,$$

where F_1 and F_2 are closed, disjoint, and non-empty. Let

$$\mathcal{O}_1 = \mathcal{U} \cap F_1 \quad \text{and} \quad \mathcal{O}_2 = \mathcal{U} \cap F_2.$$

Clearly, $\mathcal{O}_1$ and $\mathcal{O}_2$ are disjoint. If $z \in \mathcal{O}_1$, then $z \in \mathcal{U}$, and every small ball centered at z is contained in $\mathcal{U}$. If every such ball intersects F_2 , then $z \in F_2$ since F_2 is closed. However, F_1 and F_2 are disjoint, so this cannot happen. Consequently, $\mathcal{O}_1$ is open, and by the same argument, so is $\mathcal{O}_2$. Finally, we claim that $\mathcal{O}_1$ is non-empty. If not, then F_1 is entirely contained in Γ and $\mathcal{U}$ is contained in F_2 . Pick any point $z \in F_1$, which we know belongs to Γ . Now every ball centered at z intersects $\mathcal{U}$, hence F_2 . But F_2 is closed and disjoint from F_1 , so we get a contradiction. A similar argument for $\mathcal{O}_1$ proves that

$$\mathcal{U} = \mathcal{O}_1 \cup \mathcal{O}_2,$$

where $\mathcal{O}_1, \mathcal{O}_2$ are disjoint, open, and non-empty. This contradicts the fact that $\mathcal{U}$ is connected, and concludes the proof of the Jordan curve theorem for piecewise-smooth curves.

2.1 Proof of a general form of Cauchy's theorem

Theorem 2.9 *If a function f is holomorphic in an open set that contains a simple closed piecewise-smooth curve Γ and its interior, then*

$$\int_{\Gamma} f = 0.$$

Let $\mathcal{O}$ denote an open set on which f is holomorphic, and which contains Γ and its interior Ω . The idea is to construct a closed curve Λ

in Ω that is so close to Γ that $\int_{\Gamma} f = \int_{\Lambda} f$. Then, the integral on the right-hand side is 0, since f is holomorphic in the simply connected open set Ω . We build Λ as follows. Near the smooth parts of Γ , the curve Λ is essentially a curve like Γ_{ϵ} in Lemma 2.4. Near points where smooth parts of Γ join, we shall use for Λ an arc of a circle. This is illustrated in Figure 5.

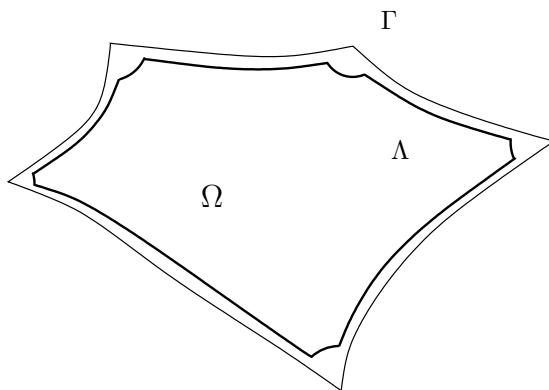


Figure 5. The curve Λ

To find the appropriate connecting arcs, we need the following preliminary result.

Lemma 2.10 *Let $\gamma : [0, 1] \rightarrow \mathbb{C}$ be a simple smooth curve. Then, for all sufficiently small $\delta > 0$ the circle C_{δ} centered at $\gamma(0)$ and of radius δ intersects γ in precisely one point.*

Proof. We may assume that $\gamma(0) = 0$. Since $\gamma(0) \neq \gamma(1)$ it is clear that for each small $\delta > 0$, the circle C_{δ} intersects γ in at least one point. If the conclusion in the lemma is false, we can find a sequence of positive δ_j going to 0, and so that the equation $|\gamma(t)| = \delta_j$ has at least two distinct solutions. The mean value theorem applied to $h(t) = |\gamma(t)|^2$ provides a sequence of positive numbers t_j so that $t_j \rightarrow 0$ and $h'(t_j) = 0$. Thus

$$\gamma'(t_j) \cdot \gamma(t_j) = 0 \quad \text{for all } j.$$

However, the curve is smooth, so

$$\gamma(t) = \gamma(0) + \gamma'(0)t + t\varphi(t) \quad \text{and} \quad \gamma'(t) = \gamma'(0) + \psi(t),$$

where $|\varphi(t)| \rightarrow 0$ and $|\psi(t)| \rightarrow 0$ as t goes to 0. Then recalling that $\gamma(0) = 0$, we find $\gamma'(t) \cdot \gamma(t) = |\gamma'(0)|^2 t + o(|t|)$. The definition of a smooth curve also requires that $\gamma'(0) \neq 0$, so the above gives

$$\gamma'(t) \cdot \gamma(t) \neq 0 \quad \text{for all small } t.$$

This is the desired contradiction.

Returning to the proof of Cauchy's theorem, choose ϵ so small that the open set $\mathcal{U}$ of all points at a distance $< \epsilon$ of Γ is contained in $\mathcal{O}$.

Next, if $P_1, \dots, P_n$ denote the consecutive points where smooth parts of Γ join, we may pick $\delta < \epsilon/10$ so small that each circle C_j centered at a point P_j and of radius δ intersects Γ in precisely two distinct points (this is possible by the previous lemma). These two points on C_j determine two arcs of circles, only one of which (denoted by Γ_j) has an interior entirely contained in Ω . To see this, it suffices to recall that if γ is a parametrization of a smooth part of Γ with end-point P_j , then for all small ϵ' the curves parametrized by $\gamma_{\epsilon'}$ and $\gamma_{-\epsilon'}$ of Lemma 2.4 lie on opposite sides of Γ and must intersect the circle C_j . By construction the disc D_j^* centered at P_j and of radius 2δ is also contained in $\mathcal{U}$, hence in $\mathcal{O}$.

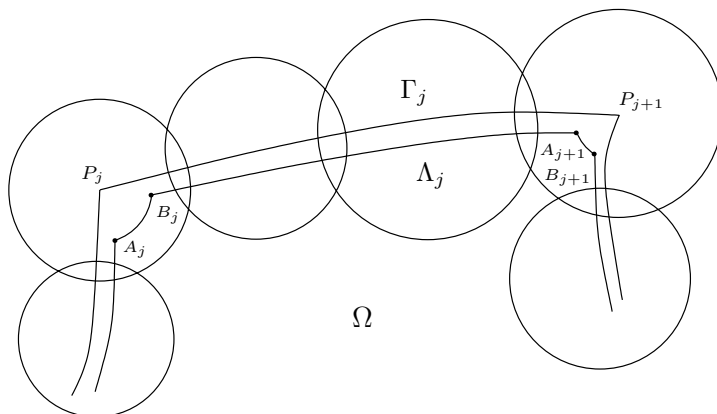


Figure 6. Construction of the curve Λ

We wish to construct Λ so that we may argue as in the proof of Theorem 5.1, Chapter 3 and establish $\int_{\Gamma} f = \int_{\Lambda} f$. To do so, we consider a chain of discs $\mathcal{D} = \{D_0, \dots, D_K\}$ contained in $\mathcal{U}$, and so that Γ is contained in their union, with $D_k \cap D_{k+1} \neq \emptyset$, $D_0 = D_K$, and with the discs

D_j^* part of the chain $\mathcal{D}$. Suppose Γ_j is the smooth part of Γ that joins P_j to P_{j+1} . By Lemma 2.4 it is possible to construct a continuous curve Λ_j that is contained in Ω and in the union of the discs, and which connects a point on B_j on $\mathcal{C}_j$ to a point on A_{j+1} on $\mathcal{C}_{j+1}$ (see Figure 6). Since we only assumed that Γ has one continuous derivative, Λ_j need not be smooth, but by approximating this continuous curve by polygonal lines if necessary, we may actually assume that Λ_j is also smooth. Then, A_{j+1} is joined to B_{j+1} by a piece of $\mathcal{C}_{j+1}$, and so on. This procedure provides a piecewise-smooth curve Λ that is closed and contained in Ω .

Since f has a primitive on each disc of the family $\mathcal{D}$, we may argue as in the proof of Theorem 5.1, Chapter 3 to find that $\int_{\Gamma} f = \int_{\Lambda} f$. Since Ω is simply connected, we have $\int_{\Lambda} f = 0$, and as a result

$$\int_{\Gamma} f = 0.$$

Notes and References

Useful references for many of the subjects treated here are Saks and Zygmund [34], Ahlfors [2], and Lang [23].

Introduction

The citation is from Riemann's dissertation [32].

Chapter 1

The citation is a free translation of a passage in Borel's book [6].

Chapter 2

The citation is a translation of an excerpt from Cauchy's memoir [7].

Results related to the natural boundaries of holomorphic functions in the unit disc can be found in Titchmarsh [36].

The construction of the universal functions in Problem 5 are due to G. D. Birkhoff and G.R. MacLane.

Chapter 3

The citation is a translation of a passage in Cauchy's memoir [8].

Problem 1 and other results related to injective holomorphic mappings (univalent functions) can be found in Duren [11].

Also, see Muskhelishvili [25] for more about the Cauchy integral introduced in Problem 5.

Chapter 4

The citation is from Wiener [40].

The argument in Exercise 1 was discovered by D. J. Newman; see [4].

The Paley-Wiener theorems appeared first in [28]; further generalizations can be found in Stein and Weiss [35].

Results related to the Borel transform (Problem 4) can be found in Boas [5].

Chapter 5

The citation is a translation from the German of a passage in a letter from K. Weierstrass to S. Kowalewskaja; see [38].

A classical reference for Nevanlinna theory is the book by R. Nevanlinna himself [27].

Chapter 6

A number of different proofs of the analytic continuation and functional equation for the zeta function can be found in Chapter 2 of Titchmarsh [37].

Chapter 7

The citation is from Hadamard [14]. Riemann's statement concerning the zeroes of the zeta function in the critical strip is a passage taken from his paper [33].

Further material related to the proof of the prime number theorem presented in the text is in Chapter 2 of Ingham [19], and Chapter 3 of Titchmarsh [37].

The “elementary” analysis of the distribution of primes (without using the analytic properties of the zeta function) was initiated by Tchebychev, and culminated in the Erdős-Selberg proof of the prime number theorem. See Chapter XXII in Hardy and Wright [17].

The results in Problems 2 and 3 can be found in Chapter 4 of Ingham [19].

For Problem 4, consult Estermann [13].

Chapter 8

The citation is from Christoffel [9].

A systematic treatment of conformal mappings is Nehari [26].

Some history related to the Riemann mapping theorem, as well as the details in Problem 7, can be found in Remmert [31].

Results related to the boundary behavior of holomorphic functions (Problem 6) are in Chapter XIV of Zygmund [41].

An introduction to the interplay between the Poincaré metric and complex analysis can be found in Ahlfors [1]. For further results on the Schwartz-Pick lemma and hyperbolicity, see Kobayashi [21].

For more on Bieberbach’s conjecture, see Chapter 2 in Duren [11] and Chapter 8 in Hayman [18].

Chapter 9

The citation is taken from Poincaré [30].

Problems 2, 3, and 4 are in Saks and Zygmund [34].

Chapter 10

The citation is from Hardy, Chapter IX in [16].

A systematic account of the theory of theta functions and Jacobi’s theory of elliptic functions is in Whittaker and Watson [39], Chapters 21 and 22.

Section 2. For more on the partition function, see Chapter XIX in Hardy and Wright [17].

Section 3. The more standard proofs of the theorems about the sum of two and four squares are in Hardy and Wright [17], Chapter XX. The approach we use was developed by Mordell and Hardy [15] to derive exact formulas for the number of representations as the sum of k squares, when $k \geq 5$. The special case $k = 8$ is in Problem 6. For $k \leq 4$ the method as given there breaks down because of the non-absolute convergence of the associated “Eisenstein series.” In our presentation we get around this difficulty by using the “forbidden” Eisenstein series. When $k = 2$, an entirely different construction is needed, and the analysis centering around $\mathcal{C}(\tau)$ is a further new aspect of this problem.

The theorem on the sum of three squares (Problem 1) is in Part I, Chapter 4 of Landau [22].

Appendix A

The citation is taken from the appendix in Airy’s article [3].

For systematic accounts of Laplace’s method, stationary phase, and the method of steepest descent, see Erdélyi [12] and Copson [10].

The more refined asymptotics of the partition function can be found in Chapter 8 of Hardy [16].

Appendix B

The citation is taken from Picard's address found in Jordan's collected works [20].

The proof of the Jordan curve theorem for piecewise-smooth curve due to Pederson [29] is an adaptation of the proof for polygonal curves which can be found in Saks and Zygmund [34].

For a proof of the Jordan theorem for continuous curves using notions of algebraic topology, see Munkres [24].

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Symbol Glossary

The page numbers on the right indicate the first time the symbol or notation is defined or used. As usual, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ denote the integers, the rationals, the reals, and the complex numbers respectively.

$\operatorname{Re}(z), \operatorname{Im}(z)$	Real and Imaginary parts	2
$\arg z$	Argument of z	4
$ z , \bar{z}$	Absolute value and complex conjugate	3, 3
$D_r(z_0), \overline{D}_r(z_0)$	Open and closed discs centered at z_0 and with radius r	5, 6
$C_r(z_0)$	Circle centered at z_0 with radius r	6
D, C	Generic disc and circle	
$\mathbb{D}$	Unit disc	6
$\Omega^c, \overline{\Omega}, \partial\Omega$	Complement, closure, and boundary of Ω	6
$\operatorname{diam}(\Omega)$	Diameter of Ω	6
$\frac{\partial}{\partial z}, \frac{\partial}{\partial \bar{z}}$	Differential operators	12
$e^z, \cos z, \sin z$	Complex exponential and trigonometric functions	14, 16
γ^-	Reverse parametrization	19
$O, o, \sim$	Bounds and asymptotic relations	24
Δ	Laplacian	27
$F(\alpha, \beta, \gamma; z)$	Hypergeometric series	28
$\operatorname{res}_z f$	Residue	75
$P_r(\gamma), \mathcal{P}_y(x)$	Poisson kernels	67, 78
$\cosh z, \sinh z$	Hyperbolic cosine and sine	81, 83
$\mathbb{S}$	Riemann sphere	89
$\log, \log_\Omega$	Logarithms	98, 99
$\hat{f}(\xi)$	Fourier transform	111
$\mathfrak{F}_a, \mathfrak{F}$	Class of functions with moderate decay in strips	113, 114
$S_a, S_{\delta, M}$	Horizontal strips	113, 160
ρ, ρ_f	Order of growth	138
E_k	Canonical factors	145
ψ_α	Blaschke factors	153

$\Gamma(s)$	Gamma function	160
$\zeta(s)$	Riemann zeta function	168
$\vartheta, \Theta(z \tau), \theta(\tau)$	Theta function	169, 284, 284
$\xi(s)$	Xi function	169
J_ν	Bessel functions	176
B_m	Bernoulli number	179
$\pi(x)$	Number of primes $\leq x$	182
$f(x) \approx g(x)$	Asymptotic relation	182
$\psi(x), \Lambda(n), \psi_1(x)$	Functions of Tchebychev	188, 189, 190
$d(n)$	Number of divisors of n	200
$\sigma_a(n)$	Sum of the a^{th} powers of divisors of n	200
$\mu(n)$	Möbius function	200
$\text{Li}(x)$	Approximation to $\pi(x)$	202
$\mathbb{H}$	Upper half-plane	208
$\text{Aut}(\Omega)$	Automorphism group of Ω	219
$\text{SL}_2(\mathbb{R})$	Special linear group	222
$\text{PSL}_2(\mathbb{R})$	Projective special linear group	223
$\text{SU}(1, 1)$	Group of fractional linear transformations	257
Λ, Λ^*	Lattice and lattice minus the origin	262, 267
$\wp$	Weierstrass elliptic function	269
$E_k(\tau), E_2^*(\tau)$	Eisenstein series	273, 305
$F(\tau), \tilde{F}(\tau)$	Forbidden Eisenstein series and its reverse	278, 305
$\Pi(z \tau)$	Triple product	286
$\eta(\tau)$	Dedekind eta function	292
$p(n)$	Partition function	293
$r_2(n)$	Number of ways n is a sum of two squares	296
$r_4(n)$	Number of ways n is a sum of four squares	297
$d_1(n), d_3(n), \sigma_1^*(n)$	Divisor functions	297, 304
$\text{Ai}(s)$	Airy function	328
$W_\gamma(z)$	Winding number	347

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